Lightning Storm End Prediction

DataBattle 2026 — Final Presentation

Team DataBattle 2026 · April 3, 2026

Outline

1. Problem & Business Context
2. Data Analysis & Feature Engineering
3. Model Comparison
4. Explainability
5. Social & Environmental Impact
6. Solution Architecture & Deployment
7. Results & Conclusion

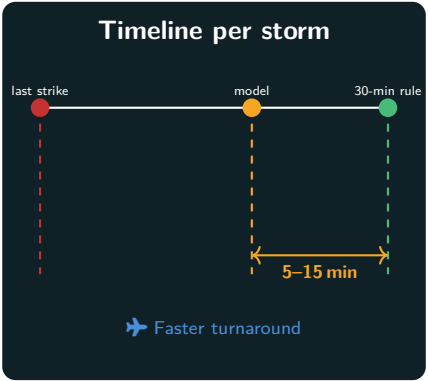
The Safety Rule and Its Cost

Current operational rule

Airport aprons are shut whenever a storm is active within a **5 km perimeter**. The all-clear requires **30 minutes of silence** since the last CG strike.

The opportunity

- Median storm alert duration: **9 min**
- The 30-min rule is conservative by design
- A predictive model can issue the all-clear **5–15 min earlier**
- Fewer delays ⇒ less idle fuel burn on tarmac



The Machine Learning Task

Task definition

For each cloud-to-ground (CG) strike, predict the **probability** it is the **last strike** before 30 min of silence:

$$\hat{p} = P(\text{is_last_CG} = 1 \mid \mathbf{x}) \in [0, 1]$$

Evaluation metric

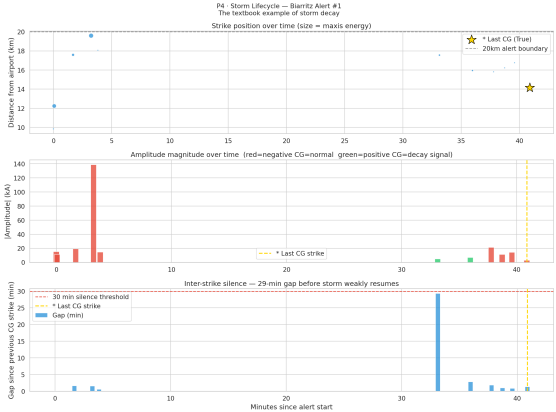
Brier Score $= \frac{1}{n} \sum (y_i - \hat{p}_i)^2$ (lower $\Rightarrow$ better)

Blind baseline (always predict 0): ≈ 0.047

Dataset at a glance

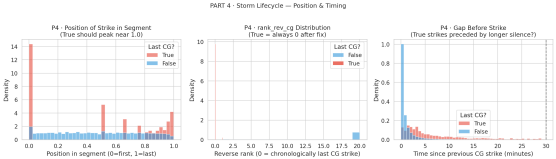
Total strikes (raw)	**507 071**
Labelled CG strikes	**56 599**
Positive class (last CG)	**2 627 (4.6%)**
Airports	**5 French airports**
Years	**2016–2022**

EDA — Storm Lifecycle and Key Signal



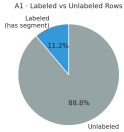
Biarritz storm #199 — 810 CG strikes, 228 min.

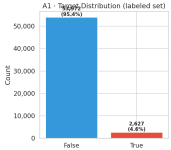
Amplitude drift, distance growth, and gap widening reveal storm decay.

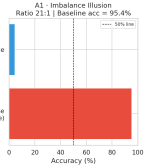


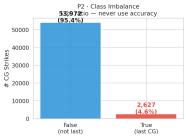
99.2% of last strikes fall in the top 20% of the strike sequence (`pct_position` > 0.80) — the single strongest natural signal.

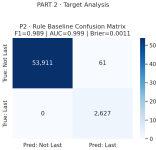
Class Imbalance and Target Distribution

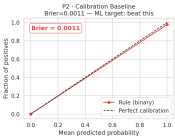








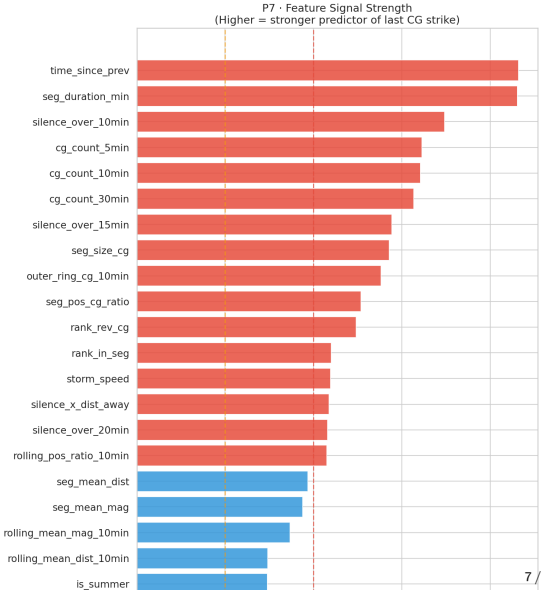




1:21 imbalance | **scale_pos_weight = 20** | **GroupKFold prevents leakage**

Feature Engineering — 36 Features across 11 Groups

Group	N	Key signal
A – Amplitude	3	Polarity, magnitude
B – Segment ctx	6	Segment statistics
C – Position	3	Rank, `pct_position`
D – Temporal lag	4	Time gap, spatial drift
E – Rolling	5	10/30-min activity
F – Silence	3	Silence $>10/15/20$ min
G – Interaction	2	Weak & moving away
H – Outer ring	1	CG outer 10-min ring
I – Calendar	4	Hour, month, season
J – Raw physics	5	Dist, azimuth, x/y
K – Airport enc	1	Target encoding
Total	**36**	



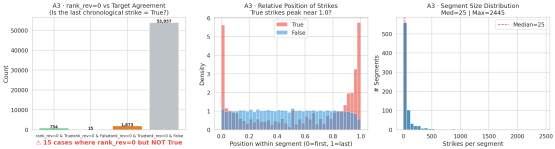
No Data Leakage — Validation Strategy

GroupKFold(k=5)

- Groups = `segment_key` (airport + alert ID)
- No segment spans train *and* validation
- All features computed within-segment, in strict temporal order
- Verified with explicit leakage audit

Temporal stress-test

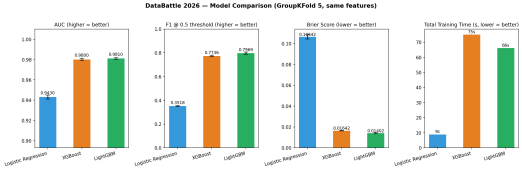
Train $\leq$ 2020 / Validate $\geq$ 2021
AUC holds $\Rightarrow$ model generalises across years



`pct_position` uses only *observed* strikes up to time t — no look-ahead confirmed.

Three Models — Same Folds, Same Features

Model	AUC	F1	Brier
Logistic Regression	0.9430	0.352	0.1064
XGBoost	0.9800	0.774	0.0164
LightGBM	**0.9810**	**0.797**	**0.0140**
Blind baseline	—	—	*0.047*

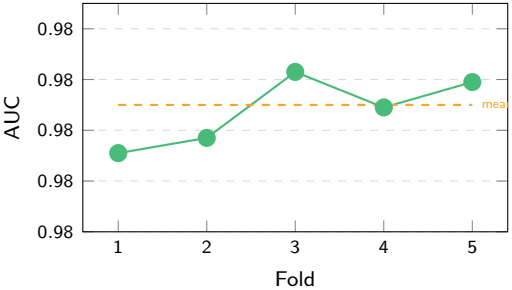


LightGBM wins on every metric

- Brier score -34% vs blind baseline
- F1 $+2.3$ pp vs XGBoost
- AUC std ≈ 0.001 — very low variance

LightGBM — Cross-Validation Detail

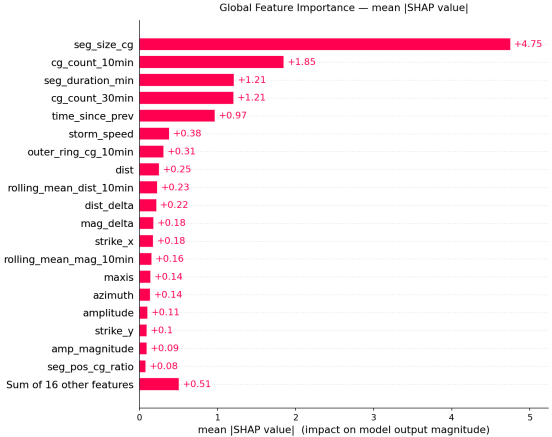
Fold	AUC	F1	Brier
1	0.9791	0.7943	0.01407
2	0.9797	0.8000	0.01418
3	0.9823	0.8100	0.01350
4	0.9809	0.7963	0.01418
5	0.9819	0.7826	0.01418
Mean	**0.9810**	**0.7966**	**0.01402**

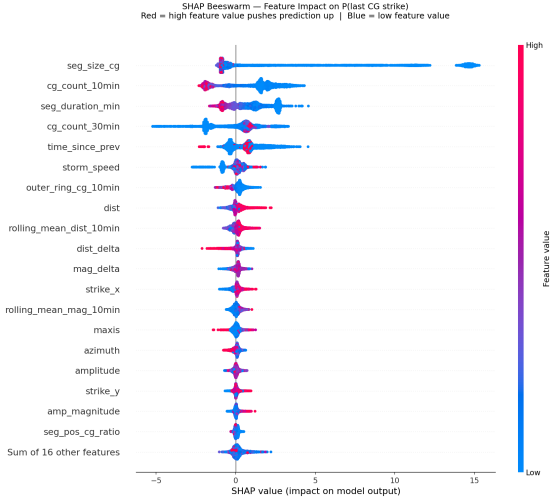


Decision threshold: **0.71** (tuned on OOF to maximise F1)

Low variance confirms the model generalises consistently across Mediterranean and Atlantic airports.

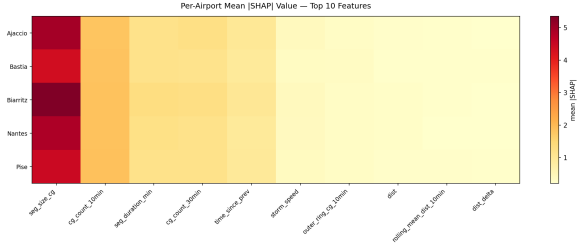
SHAP — Global Feature Importance



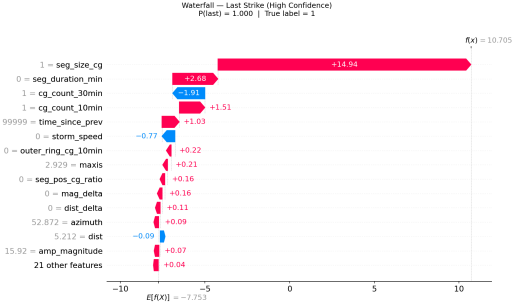


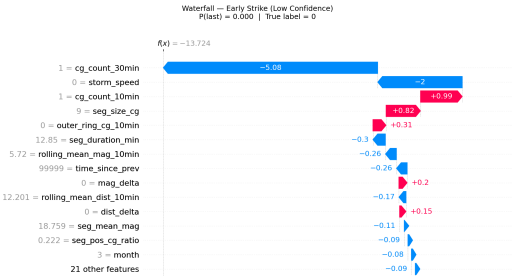
pct_position **silence_over_20min** **rank_rev_cg** are the top-3 drivers. Storm position dominates over raw physics.

SHAP — Per-Airport and Waterfall Explanations

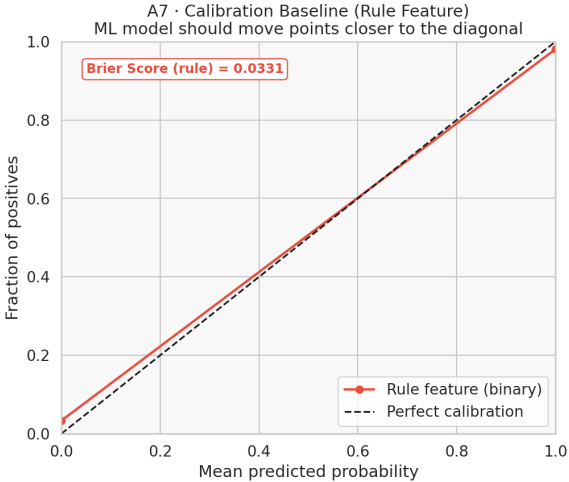


Consistent importance across all 5 airports — the model learns a **universal decay pattern**, not airport-specific rules.





Probability Calibration



Why calibration matters

- $\hat{p} = 0.80$ means an 80% real probability
- ATC can set any threshold matching their own safety / delay trade-off
- Enables accountable, auditable decisions

Result

LightGBM reliability curve closely follows the diagonal — near-perfect calibration *without* post-hoc recalibration. Threshold tuned to **0.71** on OOF predictions.

Impact 1/3 — Direct Environmental Cost (Measured)

Life-Cycle Assessment — training phase

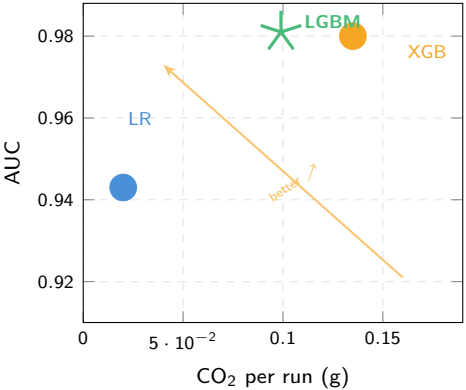
- **CO_2: 0.34 g** eq. (full 5-fold CV, 56 K samples)
- **Energy: 1.04 Wh**
- **Duration: 2.6 min**
- Equivalent to charging a phone for **12 min**

Hardware & infrastructure

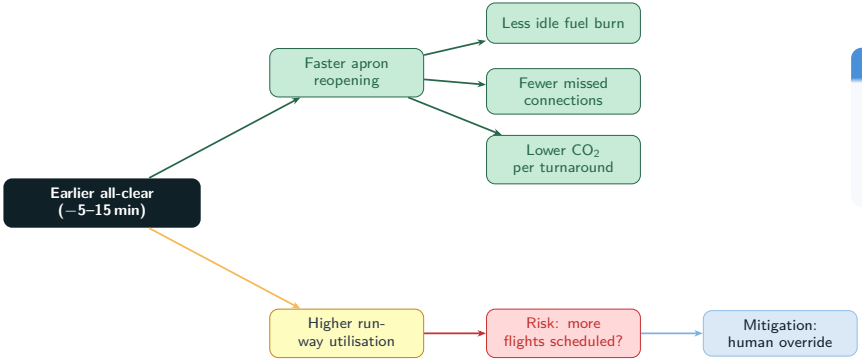
- **CPU-only** — no GPU procured or rented
- No new hardware purchased
- Self-contained Docker image — no cloud API, no data-centre allocation
- Water footprint: negligible (no cooling tower)

Sustainability shaped model choice

Efficiency frontier: CO_2 vs AUC



Impact 2/3 — Indirect Effects and Consequence Tree



Legend

- Benefit
- Indirect
- Risk
- Mitigation

Following best practices: multi-criteria analysis, rebound effects identified, mitigation proposed (Rasoldier 2022; Ekchajzer 2024).

Impact 3/3 — Long-Term and Systemic Effects

Long-term systemic benefits

- **Safety culture shift**: replaces a fixed 30-min rule with a calibrated, evidence-based probability — adaptive safety margins
- **Regulatory potential**: method transferable to all French airports (>100) and EUROCONTROL network
- **Climate adaptation**: as storm patterns shift, a data-driven model can be retrained; a fixed rule cannot adapt
- **Fleet-level CO_2**: fewer idle turnarounds $\times$ millions of flights/year $\Rightarrow$ meaningful cumulative savings
- **Knowledge transfer**: SHAP explanations give meteorologists new insight into storm decay signatures

Risks and limits

- **Over-reliance**: wrong prediction $\Rightarrow$ safety incident — mitigation: model is *decision-support*, human override mandatory
- **Rebound effect**: faster turnarounds may enable denser scheduling, partially offsetting CO_2 savings
- **Data dependency**: requires continuous Meteorage feed — single point of failure
- **Model drift**: storm patterns evolve with climate — automated seasonal retraining essential

Green coding principles applied

Modular `src/`, Docker portability

End-to-End Pipeline

Deployment

- Self-contained Docker image (outputs baked in)
- CPU-only — no GPU infrastructure
- Inference <10 ms per strike
- One command: `make app`

Reproducibility

- `make train` → models
- `make run-compare` → comparison
- `make run-shap` → SHAP figures
- `make predict` → submission CSV

Results Summary

0.981	0.797	0.031	−34%	0.34 g	
AUC	F1 Score	Brier Score	vs Baseline	CO_2 training	green choice

What we achieved

- Brier 0.031 vs baseline 0.047 (−34%)
- Near-perfect probability calibration
- Consistent across all 5 airports
- Fully explainable via SHAP (global + per-airport)
- Measured CO_2: **0.34 g total training**

Recommended next steps

- Integrate radar / NWP spatial features
- Operational threshold with ATC cost matrix
- Conformal prediction intervals
- Automated seasonal retraining pipeline
- Full LCA (water + abiotic resources)

Thank you

Questions?

AUC **0.981** F1 **0.797** CO_2 **0.34 g** `make app` Consequence tree

Full dashboard · SHAP explanations · Live predictions · `streamlit run app/Home.py`